



Original Article

A study on process information-driven cyber threat detection for I&C systems in NPP

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ABSTRACT

The increasing digitalization of Instrumentation and Control (I&C) systems in Nuclear Power Plants (NPPs) has improved operational efficiency while introducing cybersecurity vulnerabilities. Conventional network-based intrusion detection systems (IDS) face limitations in detecting sophisticated cyber threats targeting safety-critical controllers. To address these challenges, this study proposes a process information-driven cyber threat detection methodology based on real-time process data analysis and control logic consistency, enabling non-intrusive threat identification. The proposed methodology was examined through simulation and experimental testing using an APR-1400 Reactor Protection System (RPS) testbed. A cyber attack scenario targeting the High Pressurizer Pressure (HPP) Trip function was designed to assess the effectiveness of the detection mechanism. Simulation results demonstrated the detection algorithm's ability to identify unauthorized modifications to the trip setpoint, indicating the potential to detect cyber threats affecting reactor trip logic. Furthermore, experimental testing using the Safety Data Acquisition & Detection System (SDDS) demonstrated real-time anomaly detection while maintaining system integrity. These findings suggest that the proposed process-driven detection technique can enhance the cybersecurity resilience of NPPs without disrupting operational stability.

1. Introduction

The instrumentation and control (I&C) system of a nuclear power plant (NPP) serves as the central nervous system of the plant, executing critical functions related to protection, control, and monitoring, thus ensuring the safe operation of the facility [1]. Traditional I&C systems in NPPs have been designed and operated within physically isolated closed network environment, a practice that remains in place today. However, following the introduction and deployment of digital I&C systems developed through Korea's localization project over the past decade, internal connectivity among system components such as controllers, communication modules, and centralized monitoring units has significantly increased. This expanded interconnectivity, while enhancing system functionality and efficiency, has also broadened the potential cyber-attack surface, leading to new cybersecurity concerns for NPPs as safety-critical systems [2,3].

Following the Stuxnet incident, internationally recognized organizations such as the International Atomic Energy Agency (IAEA) and the U.S. National Institute of Standards and Technology (NIST) have

emphasized the importance of enhancing cybersecurity for nuclear infrastructure. They have issued technical guidelines and best-practice frameworks that include security technologies such as Intrusion Detection Systems (IDS) [4,5]. Similarly, South Korea's nuclear regulatory body, the Korea Institute of Nuclear Nonproliferation and Control (KINAC), also recommends the adoption of IDS to enhance cybersecurity in accordance with the RS-015 regulatory guideline [6]. Despite the numerous advantages offered by digital I&C systems, the increasing network connectivity has made NPPs more vulnerable to sophisticated and persistent cyber threats than ever before [7]. NPPs operate in a specialized environment with stringent requirements for safety, reliability, and availability. This complexity makes achieving adequate protection using conventional information technology (IT) security measures such as firewall-based filtering, signature-based IDS, or general network traffic analysis challenging, particularly in safety-critical and closed I&C environments like those of NPPs. Unlike conventional IT approaches that rely heavily on external traffic monitoring, the method proposed in this study leverages internal process data and control logic consistency to detect cyber threats in a non-intrusive

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manner tailored for NPP I&C systems. This approach not only has the potential to improve detection accuracy for logic-based attacks that conventional systems may overlook, but also may reduce false positives by evaluating anomalies in the context of control logic and process behavior, rather than relying solely on network patterns.

Existing cybersecurity frameworks for general IT and industrial control system (ICS) environments may not fully accommodate the distinct architectural features and operational constraints of NPP I&C systems, such as hierarchical safety structures, redundancy, and strict isolation requirements [8]. Conventional frameworks may overlook key characteristics such as redundancy, diversity, and real-time control requirements of safety-critical functions like the Reactor Protection System (RPS). While traditional network-based security detection systems, including signature-based IDS and anomaly detection systems, can still be applied in NPP environments, their effectiveness may be significantly limited due to the unique operational and regulatory characteristics of nuclear I&C systems. Therefore, a novel security approach is required to better accommodate these specific operational and regulatory requirements.

The primary objective of this study is to develop a novel cyber threat detection methodology applicable to the I&C systems of NPPs. To achieve this, we propose a new cyber threat detection framework which identifies threats through deviations from the normal operational behavior of NPP I&C systems. Unlike conventional approaches, this study employs a detection strategy grounded in the unique process characteristics of nuclear I&C systems. The methodology proposed targets the Bistable Processor (BP) controller within the RPS. This approach aims to detect sophisticated cyberattacks against PLC controllers while minimizing false positives.

2. Background and related works

Traditional cybersecurity detection techniques used in information technology (IT) and operational technology (OT) systems encompass signature-based detection, anomaly detection, and whitelist-based detection [9]. While each technique has strengths, they also have inherent limitations.

Signature-based detection can swiftly and accurately identify known attack patterns; however, it struggles to effectively detect new types of zero-day attacks or modified attack patterns [10]. Anomaly detection learns normal system behavior and detects deviations, making it effective for identifying new types of attacks. However, in complex operational environments like NPPs, this approach may lead to a higher false positive rate [11]. Whitelist-based detection enhances security by permitting only predefined actions; yet it can pose significant maintenance challenges in complex environments like NPP I&C systems [12].

Recently, machine learning (ML) and artificial intelligence (AI)-based approaches for cyber threat detection in ICS have attracted significant attention. Zhao et al. [13] proposed a method to detect anomalies in ICS using a one-dimensional convolutional neural network (1D-CNN) and a bidirectional long short-term memory (BiLSTM) network. Zhang and Coble [14] developed a technique to detect false data injection attacks (FDIA) employing auto-associative kernel regression (AAKR) and support vector regression (SVR) models with a hardware-in-the-loop (HIL) testbed. Yang et al. [15] integrated AAKR with a statistical probability ratio test (SPRT) to propose an anomaly detection method for SCADA systems. Kundap et al. [16] conducted a study training an XGBoost-based anomaly detection model through HIL simulations. Haider et al. [17] introduced a machine learning-based detection technique leveraging real-time network data analysis, while Das et al. [18] proposed a logic-based anomaly detection (LAD) approach capable of real-time detection even in resource-constrained environments.

2.1. Limitations of applying conventional detection techniques in the NPP environment

As discussed in Section 1, the NPP environment is a specialized operational setting that requires high reliability and availability, which limits the applicability of conventional IT and ICS security techniques [19]. NPP I&C systems cannot afford downtime, and frequent security patches are restricted, rendering traditional signature-based detection methods less effective. Additionally, as NPP systems maintain a closed network structure with limited connectivity to external networks, it is practically difficult to implement large-scale data collection and real-time analysis essential for ML and AI-based detection techniques.

Given these operational characteristics, conventional security detection techniques face the following limitations when applied to NPP I&C systems.

- Signature-based detection: Requires periodic pattern updates, but applying regular security patches in NPP I&C systems is challenging, making it difficult to integrate the latest threat intelligence.
- Anomaly detection: The normal operational patterns of NPP systems vary considerably with fuel cycles, load conditions, and environmental changes, often leading to a high false positive rate.
- Whitelist-based detection: Permits only predefined normal operations, making system modifications inflexible and complicating maintenance.
- Machine learning-based detection: Necessitates extensive data to learn normal and abnormal patterns, but acquiring sufficient operational data in an NPP environment remains a challenge.

Given these limitations, implementing conventional cyber threat detection techniques in the NPP environment requires a tailored approach that considers the unique operational characteristics of NPPs.

2.2. Structure and security focus of control systems in NPP I&C architectures

The I&C system of an NPP has a multi-layered structure, with each layer having distinct security requirements. The primary layers include the field device layer, comprising sensors and actuators; the control layer, which is responsible for executing control logic; and the operator interface layer, where operators monitor and manage plant operations [20].

While this multi-layered structure is essential for overall operation, this study focuses primarily on the control layer, which plays a critical role in executing automated control functions and maintaining plant safety and reliability. Fig. 1 depicts the schematic structure of the APR-1400 digital I&C system, which represents the most recent NPP model in South Korea. The Man-Machine Interface System (MMIS) of the APR-1400 integrates the I&C system with the Human-Machine Interface (HMI) in a network-based distributed control architecture, enhancing functionality and operational efficiency. In this structure, the control layer components including Programmable Logic Controllers (PLCs) and Distributed Control Systems (DCS) are particularly critical for plant operation, as they manage important control logic and real-time decision-making processes. This hierarchical structure is designed to allow each layer to function independently yet interconnects seamlessly with the others. Notably, the principles of defense-in-depth, redundancy, and diversity are incorporated to maximize the system reliability and safety.

The I&C systems of the APR-1400 are broadly classified into safety-related and non-safety-related systems. Safety-related systems include the Reactor Protection System (RPS) and Engineering Safety Feature Component Control System (ESF-CCS), which perform the plant's core safety functions. Non-safety class systems include Nuclear Steam Supply System (NSSS) control systems and plant monitoring and diagnostic systems, primarily supporting the operational efficiency and

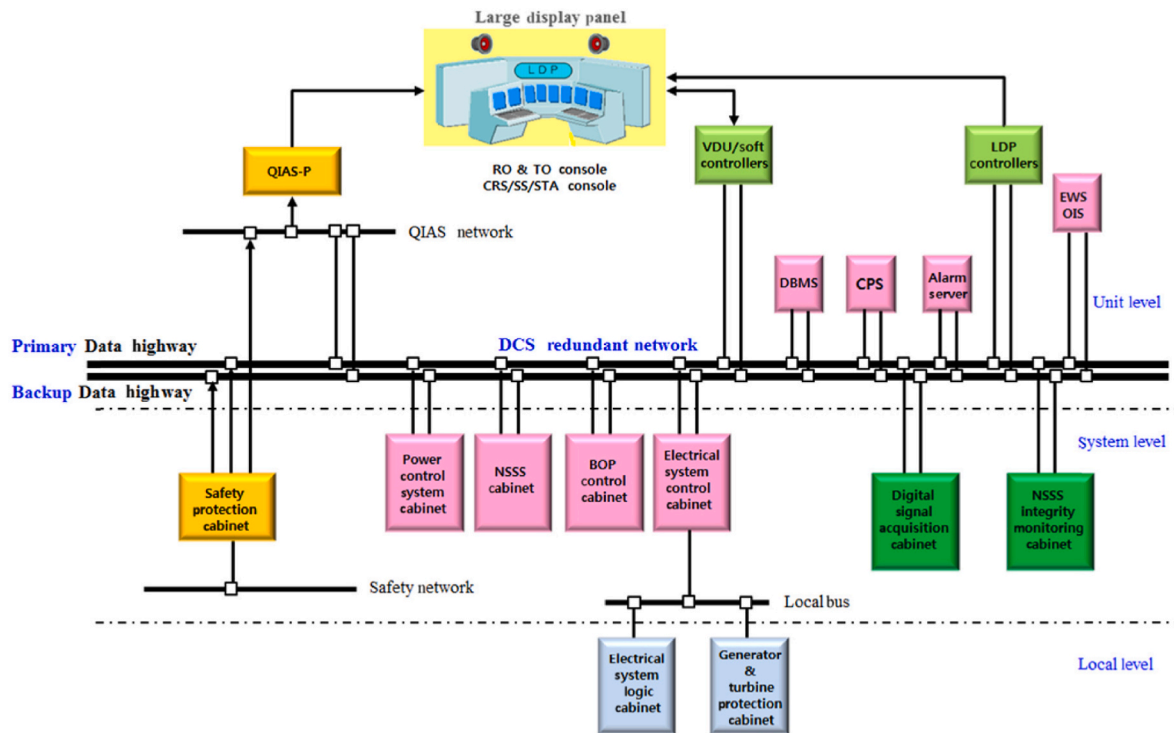


Fig. 1. Schematic structure of the APR-1400 I&C system [21].

maintenance of the plant. By clarifying the system structure and focusing specifically on the control layer, this study aims to develop a cybersecurity approach tailored to the unique operational characteristics and security challenges of critical control components within the NPP I&C environment.

3. Proposed methodology

3.1. Non-invasive data acquisition for cyber threat detection

To effectively detect cyber threats originating from controllers in national infrastructure such as NPPs, a data collection method that preserves reliability and real-time performance is essential [22]. Existing network-based detection techniques identify threats through traffic analysis, however in an NPP environment, the increased network load and the impact on real-time operations must be considered. Therefore, this study proposes applying non-invasive data collection techniques to effectively acquire security data without disrupting control system operation.

The data acquisition method applied in this study employs a network Test Access Point (TAP) to replicate the data transmitted and received from the controller network. The data acquisition device is designed to collect data in real time without altering the flow of control signals and operates independently without requiring external network connectivity. Notably, the unidirectional data collection structure allows the detection system to receive data from the NPP control network, while preventing any transmission back to the control network, thus enhancing security. This non-invasive data collection approach establishes a foundation for effectively detecting cyber threats while preserving control system safety.

Data communication in the RPS utilizes the HR-SDN network protocol and the HR-SDL data link communication method based on Profibus-FDL [23]. Therefore, as shown in Fig. 2, network data TAP devices (data mirroring devices) can be installed on the input and output modules of each controller to collect process data transmitted over the network for use in cyber threat detection. The TAP devices used here are

specially designed unidirectional network TAPs that only permit data reception, ensuring that normal controller operations remain unaffected [24]. These unidirectional network TAP devices are designed to be unable to transmit data and can completely block the impact of a detection device on the control system network. The network frame data collected through these TAP devices can be transformed and remapped to the respective process variables, which can then be utilized as input variables for detection logic based on the controller and network configuration.

In order to effectively analyze the specific information contained in frame data, it is essential to understand the structure of the frame and the meaning of each field means [25]. A frame refers to a unit of data transmitted at the data link layer in a communication network and contains various types of information depending on the communication protocol used. Also, the organization and format of frames depends on the communication protocol used.

With a basic understanding of the structure of data frame and communication protocols used in the RPS controllers, it is possible to selectively collect and process only the data needed for cyber threat detection. However, fully understanding the structure and semantics of these frames can only be done by network architects, developers, or experts with in-depth knowledge of the system [26].

While some data can be inferred and interpreted from standard protocols such as Ethernet, IP, TCP, and UDP, fully analyzing the meaning of a particular piece of data is a practical challenge [27]. However, with a clear understanding of the structure and composition of frames, it is possible to effectively distinguish between normal frame characteristics and abnormal patterns, which can significantly improve the accuracy of detecting abnormal conditions or cyber threats in controllers [28]. Understanding frame structure can also reduce unnecessary data analysis and focus on areas of interest, further improving detection processing speed [29].

Knowledge of frame structure also allows you to apply customized rules to detect specific types of attacks or anomalous behavior, which allows for more flexibility in aligning with security policies. On the other hand, when trying to detect cyber threats using ML/AI techniques

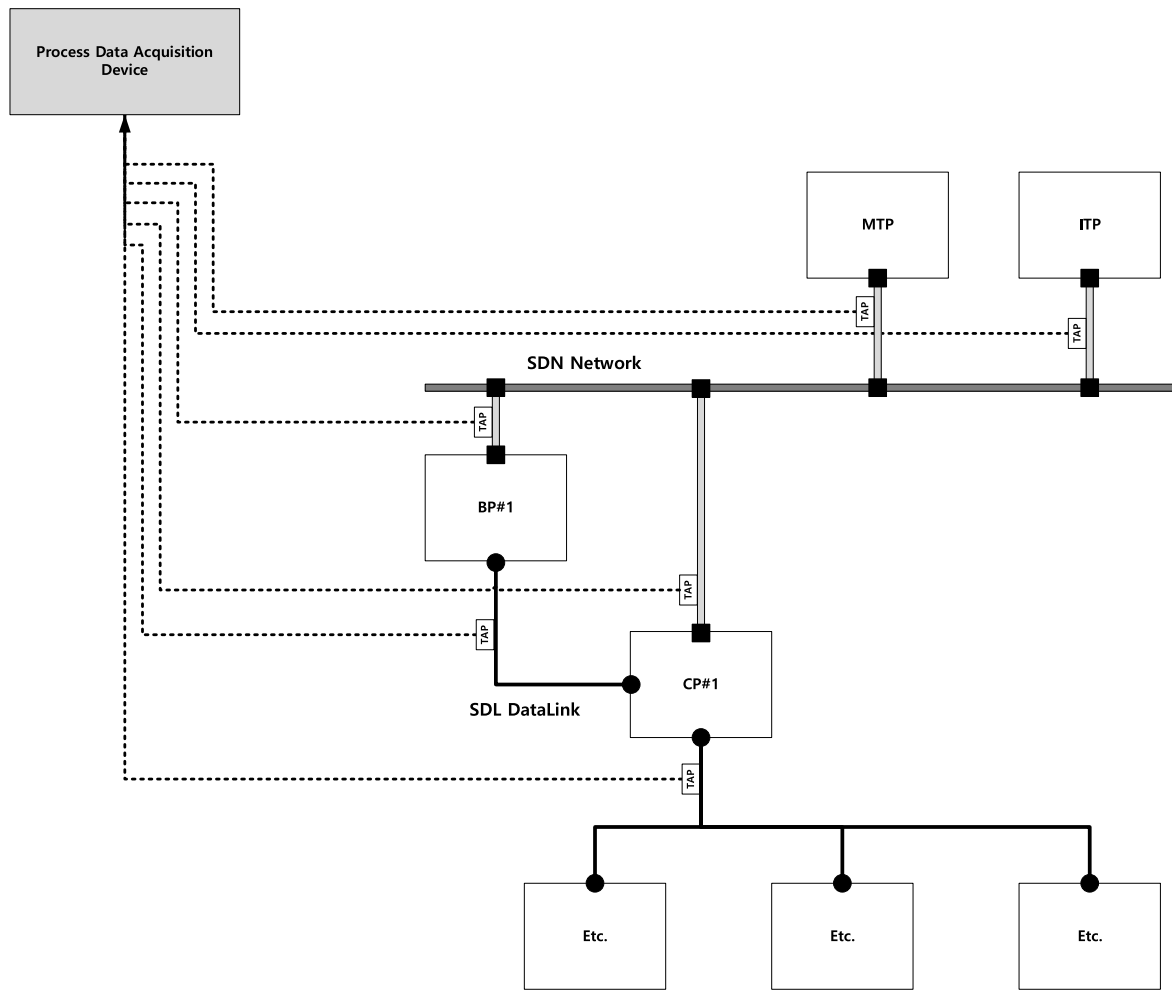


Fig. 2. Acquisition strategy for process data in detecting cyber threats to the RPS.

without understanding the structure and composition of frames, it becomes difficult to determine which features are important or which behavior is anomalous [30]. This increases the likelihood of learning the wrong patterns or over-relying on unimportant features.

3.2. Design of the detection algorithm

In this study, a cyber threat detection algorithm was developed leveraging the process data of the control system, the control logic implemented in the controller, and the health diagnosis information of the controller. The core concept of the proposed detection algorithm is verifying whether the reactor trip function of the RPS operates as intended. For this purpose, the detection targets the High Pressurizer Pressure (HPP) trip function by defining the normal operation criteria of the system and comparing these criteria with the control logic.

The design of the detection algorithm follows the following procedure.

- Establish criteria based on system design information and control logic: Identify the process variables required to perform the main functions of the RPS and define the criteria for normal operation.
- Set invariant detection rules: Utilize setpoints, exception handling rules, and controller diagnostic information. The rules assess the consistency between control logic execution and controller health status to identify potential cyber threats.

- Threat detection through real-time comparison: Determine the cyber threat status by comparing steady-state process variables and setpoints in real time.

Furthermore, to enhance the robustness of threat detection, the proposed methodology monitors not only process values and setpoints but also diagnostic signals provided by PLC controllers. These signals reflect the internal health status of the controllers. Discrepancies where a controller reports a normal diagnostic status but deviates from pre-defined control logic are interpreted as cyber threats, while abnormal diagnostic signals accompanying abnormal behavior are treated as system malfunctions. Fig. 3 illustrates a cyber threat detection logic diagram for a typical reactor trip signal generation function of an RPS. The detection logic verifies whether the Bistable Logic Processor (BP) controller is properly generating the reactor trip signal. To do this, the detection algorithm utilizes process data, setpoint verification, and test and health diagnostic signals to identify cyber threats. First, process data such as pressure, water level, temperature, flow rate, and reactor power signals are collected in real time and compared with setpoints to verify the proper operation of the reactor trip logic. This approach effectively detects anomalies such as setpoint tampering and control logic alterations. Additionally, detection accuracy is enhanced by leveraging test-related signals and diagnostic status signals to determine whether the controller is in maintenance or test mode.

Based on this detection logic, the algorithm verifies whether the PLC controller operates according to the intended logic under normal conditions and flags it as a threat if the reactor trip logic output deviates

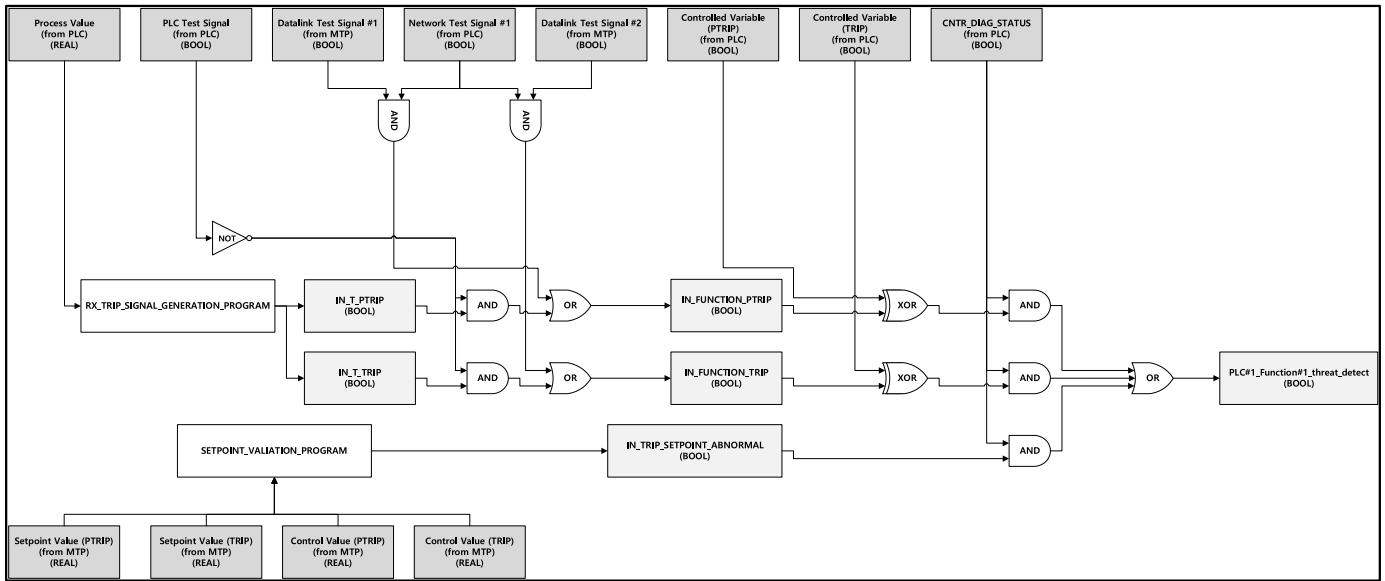


Fig. 3. Typical cyber threat detection logic diagram for TRIP generation function in RPS.

from the expected value despite normal controller operation. The definitions and roles of the variables used in the detection logic diagram are summarized in Table 1.

Among the various protective functions of the RPS, the HPP Trip function plays a key role in protecting the reactor from over-pressurization conditions. If this function is compromised, it poses a serious safety risk. Therefore, this study designed an algorithm to detect cyber threats targeting the HPP Trip logic.

The core logic of the HPP trip function is as follows. This function employs protection logic that safely shut down the reactor if the measured pressurizer pressure exceeds a predefined threshold. First, if the PZR pressure rises above the pre-trip setpoint, the HPP pre-trip output signal is set to TRUE. Conversely, if the pressure decreases below the pre-trip setpoint minus the hysteresis value, the pre-trip signal changes to FALSE. Similarly, when the pressurizer pressure exceeds the trip setpoint, the HPP trip output signal is set to TRUE; it resets to FALSE when the pressure falls below the trip setpoint minus the hysteresis value.

In this study, three detection rules were established for cyber threat detection targeting the HPP trip function of the RPS.

- **Setpoint Integrity Check:** The detection system shall compare, in real time, each setpoint value received from the controller with the corresponding value maintained internally within the detection logic, using a dedicated setpoint verification program. If any mismatch is observed (e.g., altered numerical value), the rule is violated, indicating a potential cyber threat compromising the reactor trip functionality.
- **Trip Signal Verification:** The detection system shall monitor, in real time, whether the reactor trip signal is correctly generated when the process pressure value reaches or exceeds the predefined setpoint according to the controller's logic. If no trip signal is observed despite the process variable exceeding the setpoint, the rule is violated, indicating a potential cyber threat compromising the control logic.
- **Cyber Threat Detection Condition:** If the controller's diagnostic signals are normal, but Rule 1 or Rule 2 are violated, it is judged to be control setpoint tampering or control logic alteration. If these anomalies are detected, it constitutes a cyber threat condition, triggering an alarm. If these anomalies are detected, it constitutes a cyber threat condition, triggering an alarm. This multi-layered validation approach, which combines process

variable monitoring, setpoint verification, and diagnostic signal analysis, enhances the robustness of cyber threat detection by reducing the likelihood of misinterpretation due to hardware faults or malfunctions.

3.3. Simulation-based cyber threat detection modeling and analysis

To evaluate the effectiveness of the proposed process information-driven cyber threat detection method, a simulation-based model was developed to reproduce realistic cyberattack scenarios targeting the HPP trip function of the RPS. This simulation not only validates the algorithm's detection capability but also demonstrates its performance under specific attack conditions, helping illustrate how the detection rules operates in practice. For this purpose, a simplified test model was constructed using MathWorks' Simulink tool [31].

Fig. 4 illustrates the simplified cyber threat test model, comprising the RPS model and the cyber threat detection system. The blocks on the left represent initial values and process variables supplied to the system. Initial values include major control parameters, while process variables are real-time simulated data modeled after the RPS behavior. The "RPS_TYPICAL_TRIP_LOGIC" block implements a simplified version of the RPS trip signal generation logic, triggering pre-trip and trip signals when process variables exceed predefined setpoints. The "RPS_TYPICAL_TRIP_ABNORMAL_DETECT" block monitors the control system's behavior in real time to detect deviations indicative of cyber threats.

To validate the performance of the proposed detection algorithm, simulation tests were conducted on a scenario where the setpoint was maliciously altered within the control logic. In the test scenario, a process was designed in which the PZR pressure gradually exceeds the trip setpoint to simulate a realistic situation. As shown on the left in Fig. 5, the simulation scenario is structured so that the pressure gradually increases over time. The following hypothetical setpoints were applied for this test:

- High PZR Pressure Pre-Trip Setpoint: Assumed to be 161.55 kg/cm²A
- High PZR Pressure Trip Setpoint: Assumed to be 167.55 kg/cm²A

The setpoints used in this test are assumed based on NPP design specifications. Furthermore, since this test focuses not on the absolute setpoints themselves but on detecting deviations from normal controller behavior using the proposed detection algorithm, the assumed setpoints

Table 1
Definition and role of variables in the proposed detection system.

Type	Variable name	Description
Input variables	Process Value	This is the measured process value used to determine whether the reactor trip should be triggered based on the corresponding input variable. It represents the actual measured value provided by the field instrument to the target PLC controller through hardwired connections. This signal is transmitted from the PLC to the Maintenance Test Panel (MTP) via the network and is acquired through a TAP device.
	Setpoint Value (PTRIP)	This is the setpoint information provided from the MTP to the target PLC controller.
	Setpoint Value (TRIP)	These values are transmitted from the MTP to the PLC controller via the network and are acquired through a TAP device. They are used to verify the integrity of the setpoint information stored in the target controller.
	Control Value (PTRIP), Control Value (TRIP)	This is a fixed value used to verify the integrity of the setpoints in the target controller and is stored in the internal memory of the detection system.
	PLC Test Signal	This is used to determine whether the controller is in testing mode. These values are signals transmitted from the PLC controller to the MTP via the network and are acquired through a TAP device.
	Network Test Signal #1, Datalink Test Signal #1, #2	This is used to determine whether the communication module is in testing mode. It is a signal provided from the MTP to the target controller and is acquired through a TAP device.
	Controlled Variable (PTRIP)	This is used to verify whether the reactor trip output signal from the controller is being transmitted correctly.
	Controlled Variable (TRIP)	It is a signal provided from the controller to the MTP and is acquired through a TAP device.
	CNTR_DIAG_STATUS	This is used to determine whether the controller is in a normal state or a faulty state. It is a signal provided from the controller to the MTP and is acquired through a TAP device.
	IN_T_PTRIP	This is a variable generated within the detection logic, calculated based on the input Process Value and Setpoint Value.
Internal variables	IN_T_TRIP	It uses the same logic as the reactor trip logic of the corresponding PLC controller to generate Pre-trip and Trip results.
	IN_TRIP_SETPOINT_ABNORMAL	Generates a result by comparing the setpoint value stored in the internal memory of the detection logic with the setpoint value provided by the corresponding PLC controller. If there is a discrepancy between the two values, it detects an abnormal state of the setpoint and generates a result accordingly.
	IN_FUNCTION_PTRIP IN_FUNCTION_TRIP	This is a variable generated within the detection logic, used to distinguish whether the IN_T_PTRIP and IN_T_TRIP values occurred during normal operation or during controller and network testing.
Output Variables	PLC#1_Function#1_threat_detect	Generated by comparing the real-time output value of a specific reactor trip function in the PLC controller with the result generated by the detection logic. If a discrepancy occurs between the output value and the detection logic result, or if an incorrect setpoint is entered, it is identified as a cyber threat.

have no impact on detection performance or the test's main objective.

As shown on the left in Fig. 5, under normal operating conditions, the pre-trip signal and trip signal should be activated when the PZR pressure exceeds the setpoint. Specifically, the pre-trip signal should be activated when the PZR pressure exceeds the HPP pre-trip setpoint (HPP_PTSP) of 161.55 kg/cm²A. If the pressure continues to rise and exceeds the HPP trip setpoint (HPP_TSP) of 167.55 kg/cm²A, the reactor trip signal shall be triggered. To illustrate this behavior and evaluate the detection logic under attack conditions, the following representative test case was designed and analyzed.

3.3.1. Test case: interrupting reactor trip activation by tampering with the HPP trip setpoint

This test is intended to evaluate how accurately a cyber threat detection system can identify when the trip setpoint in the HPP trip function of a RPS has been tampered with.

The time series graph on the left in Fig. 6 shows a scenario where the process variable (PZR_PRSS_NR) gradually increases after about 500 s, exceeding the existing trip setpoint. Under normal circumstances, this condition should have generated a reactor trip signal, but the setpoint was tampered with at the 500-s mark, intentionally changing the trip setpoint to interrupt a trip signal from occurring.

Fig. 7 shows the results of the detection algorithm's successful detection of an abnormal condition in the test scenario where the HPP trip setpoint was tampered with at the 500-s mark to maliciously prevent a reactor trip signal from occurring. This result confirms that the detection algorithm effectively identifies abnormal conditions in the RPS resulting from trip setpoint tampering.

4. Hardware implementation and experimental validation

The APR-1400 digital I&C system testbed was developed to facilitate a comprehensive cybersecurity study of NPP digital I&C systems [32].

The testbed comprises actual hardware controllers identical to those used in the APR-1400 plant to ensure consistency at the hardware and system levels. Although it does not replicate the full operational environment, it maintains critical hardware characteristics necessary for cybersecurity evaluation.

As shown in Fig. 8, the testbed consists of safety-critical systems and non-safety systems configured within a single channel. Safety-critical functions, such as those of the Reactor Protection System (RPS) and the Engineered Safety Feature Component Control System (ESF-CCS), are implemented using POSAFE-Q safety-class PLCs certified for nuclear applications.

This configuration enables realistic experimental evaluation of cybersecurity measures, resilience testing, and vulnerability analysis for NPP control systems. It provides an optimal platform for validating process information-based cyber threat detection methodologies under conditions that closely reflect actual plant environments. To implement and validate the proposed process-based cyber threat detection methodology in a realistic NPP I&C system environment, an independent hardware system, the Safety Data Acquisition & Detection System (SDDS), was developed and integrated with the testbed.

To validate the proposed cyber threat detection technique using a testbed environment that closely mirrors an actual NPP I&C system, a hardware system named the Safety Data Acquisition & Detection System (SDDS) was designed and developed.

SDDS is an independent hardware system that utilizes a non-invasive network TAP method to collect communication data from safety-grade PLC controllers while ensuring no interference with their operation. Unlike traditional network-based security solutions, SDDS focuses on analyzing process data from PLC controllers and detecting cyber threats in real time using its built-in detection engine. The SDDS hardware consists of 14 slots, including one CPU module and up to 13 Safety Data Network (SDN) and Safety Data Link (SDL) communication modules, making it suitable for safety-critical systems such as the RPS. Fig. 9

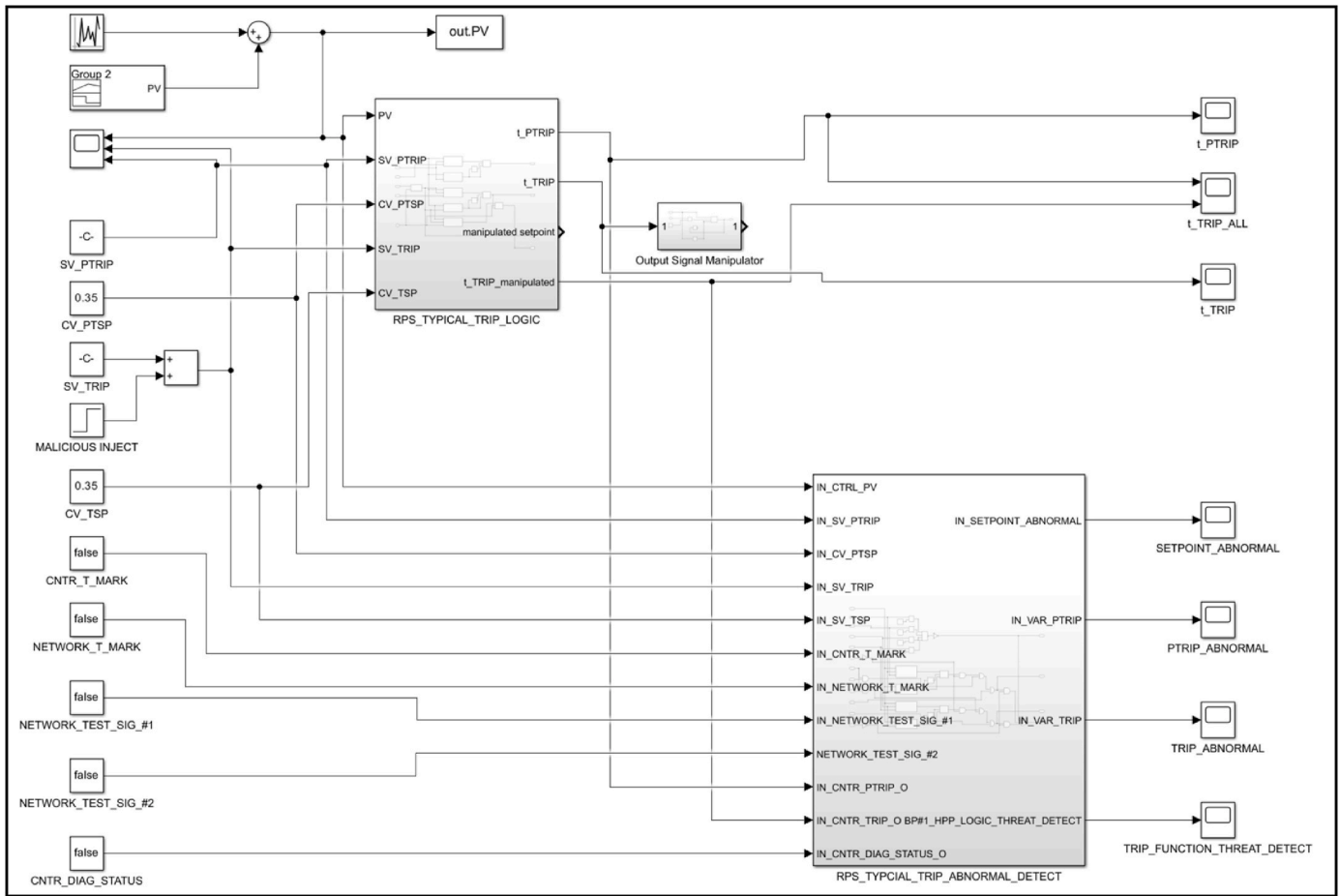


Fig. 4. Overview of threat detection system simulation modeling for attack scenario testing.

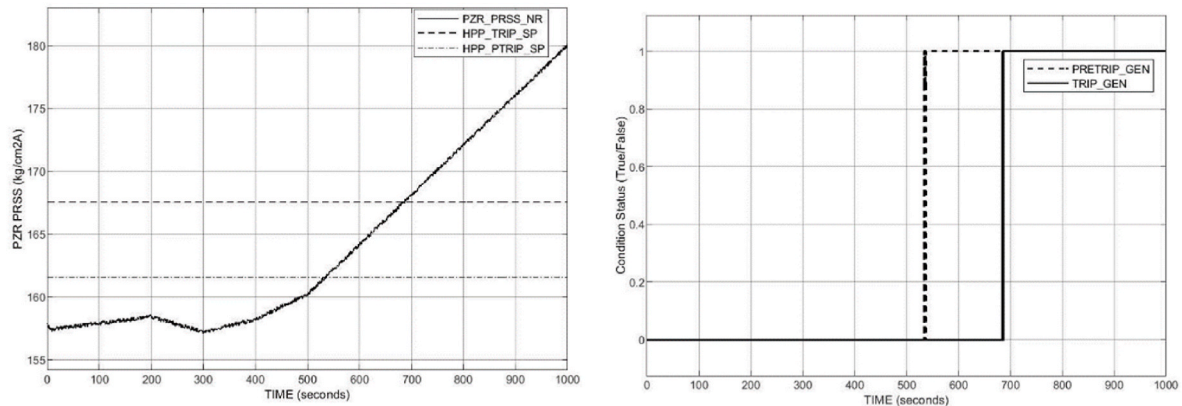


Fig. 5. Progression of PZR pressure exceeding trip setpoint (left), activation of pre-trip and trip signals relative to setpoint exceedance in PZR pressure (right).

shows its physical configuration.

The main components of SDDS are as follows:

- Power Module: Provides reliable power to the entire system.
- CPU Module: Contains an ARM-based embedded processor to run detection algorithms and perform real-time data analysis.
- SDN Module: It collects SDN (Safety Data Network) communication data via Profibus RS-485 via DSUB-9 connector and is designed as a Receive-Only interface so that it does not affect the control network.
- SDL Module: Supports optical input terminals and pass-through output terminals to monitor Rx and Tx ports for optical-based data reception and provides a data collection method suitable for the Safety Data Link (SDL) protocol used in NPP safety system control networks.
- Ethernet Interface: Provided for connection with external hosts and data analysis servers and provides scalability to transmit cyber threat detection results generated by the detection engine to other systems.

SDDS utilizes network TAP devices installed on communication

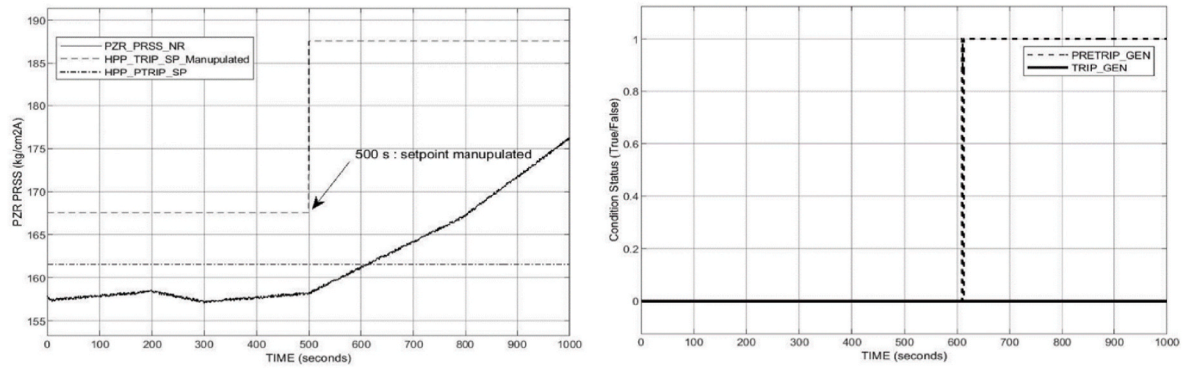


Fig. 6. Manipulation of trip setpoint to prevent reactor trip despite exceeding critical variables setpoint (left), reactor trip generation status of RPS (right).

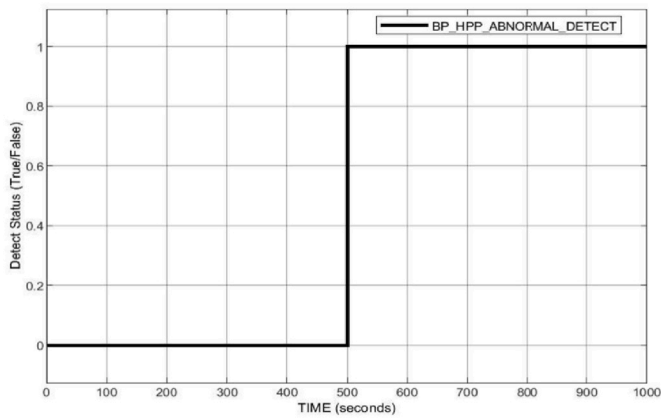


Fig. 7. The result of threat detection in case of manipulating trip setpoint to suppress reactor trip.

cables connected to safety-rated PLC controllers in the RPS to passively capture SDN and SDL data without data loss. The collected frames are analyzed in real time by the detection engine implemented in the CPU module, which determines whether critical PLC control functions have been tampered with. This approach ensures non-intrusive monitoring while enabling real-time cyber threat detection.

Fig. 10 illustrates the linkage between SDDS's process-based cyber threat detection capabilities. SDDS collects data from the NPP I&C system through the SDN and SDL frame processing modules and analyzes process variables in real time on each PLC-specific detection module to determine abnormal behavior and tampering of control logic. Detected threat information is logged and reported through the threat detection data logging and transmission module, while the system diagnostics

module monitors the status of the SDDS hardware and detection engine.

Fig. 11 illustrates the system linkage diagram for validating cyber threat detection capabilities in conjunction with the SDDS and the RPS testbed. The I/O simulator provides input signals via hardwired process signal inputs to the PLCs under test (Testbed PLC #1 and #2), each of which generates trip signals to perform safety functions of the RPS. SDDS collects SDN and SDL data through the network TAP to monitor communication between PLCs and executes detection logic based on it to analyze abnormalities.

Fig. 12 illustrates the actual test device configuration of the SDDS in conjunction with the RPS testbed. The RPS testbed comprises the RPS Maintenance Test Panel (RPS-MTP), the RPS Bistable Processor (RPS-BP), the RPS Coincidence Processor (RPS-CP), and the SDDS. The SDDS collects communication data between the RPS-BP, RPS-CP, and RPS-MTP through TAP devices and executes detection logic to identify cyber threats. As shown in the figure, the SDDS is directly connected to the network between the RPS-BP, RPS-CP, and RPS-MTP through TAP device, and is configured to provide a UI screen for testing and visualization of the results through the RPS-MTP.

To validate the cyber threat detection capability of the developed SDDS for the HPP trip function, a test was conducted by intentionally tampering with control settings. To verify that the RPS trip logic was operating normally, the safety grade PLC controller EWS (Engineering Workstation) and pSET (PLC Controller Development Tool) were utilized to manipulate setpoints in specific comparative FBD blocks to simulate a situation where the HPP trip signal would not be generated normally. Under these modified conditions, the SDDS successfully detected the behavior of the modified controller and detected abnormal behavior resulting from setpoint modification, as demonstrated in Fig. 13 and the simulation test. Cyber threat detection results of HPP Logic Tampering Log records other than the log records were blurred for RPS design information security reasons.



Fig. 8. APR-1400 digital I&C system test-bed equipment for research purposes.



Fig. 9. Physical configuration of cyber threat detection system hardware (SDDS).

5. Discussion

In this study, a process information-driven cyber threat detection method was proposed for NPP I&C systems and experimentally verified its feasibility by simulating the HPP trip function of the RPS using simulation tools and by developing actual hardware linked to the testbed, incorporating the safety-grade PLC controller used in APR-1400 NPPs.

The SDDS developed in this study successfully detected cyber threats, including setpoint manipulation. Unlike existing detection techniques based on network traffic analysis, the proposed method leverages internal process data from PLC controllers to detect cyber threats, confirming that more reliable detection is achievable by comparing threats against normal control logic.

Existing research on ICS cybersecurity, including NPPs, has limited experiments conducted using testbeds that closely resemble actual NPP I&C environments, and minimal research has been conducted on safety-rated PLC controllers in NPPs. This research differs from previous studies as it directly employs safety-rated PLC controllers and control networks (SDN, SDL) utilized in the APR-1400 NPP environment. In addition, unlike existing network traffic analysis-based detection techniques, this study proposed a cyber threat detection strategy that can operate independently of the control system even in a network isolation

environment by utilizing process information driven detection techniques such as control logic and process variables.

Although the proposed methodology demonstrated significant results through simulation and testbed trials, several limitations. First, this study focused solely on the HPP trip function, a subset of the RPS, and its applicability to other major I&C systems, such as reactor power control and feedwater control, remains unexplored. Future research should extend the methodology to various control systems and control functions. Second, this detection method relies on specific environmental and process-based knowledge of the nuclear plant, such as control logic, communication frame information, network structure, etc. Finally, as NPP I&C systems must comply with stringent regulatory requirements, the cyber threat detection system proposed in this study requires additional certification and validation before deployment in a real plant.

To compensate for the limitations identified in this study, the following additional research efforts are recommended for future work. First, it is necessary to expand the scope of detection to various I&C such as Reactor Regulating System (RRS) and Feed Water Control System (FWCS) and etc. In addition, it is necessary to optimize the detection algorithm by quantitatively evaluating the performance such as false positive rates and false negative rates, detection speed, and system load.

A key contribution of this research is the development and experimental validation of a cyber threat detection technique and system

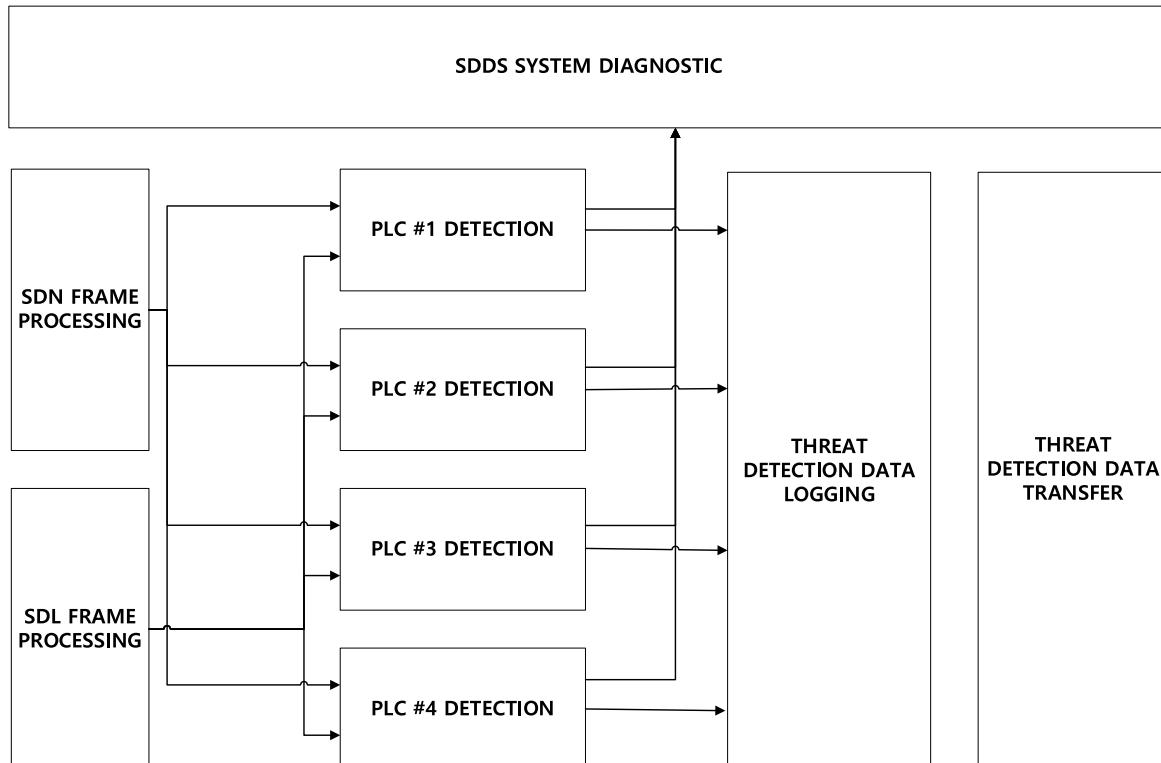


Fig. 10. Process-based cyber threat detection function diagram.

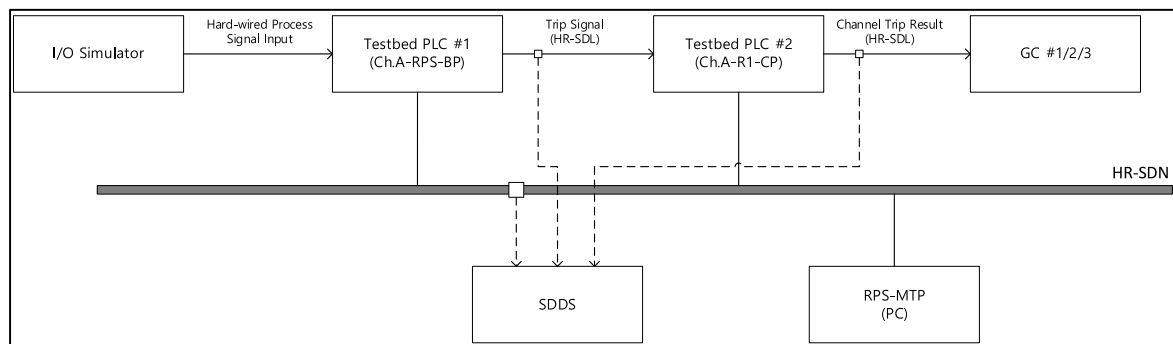


Fig. 11. Cyber threat detection system integration diagram linked with the testbed.

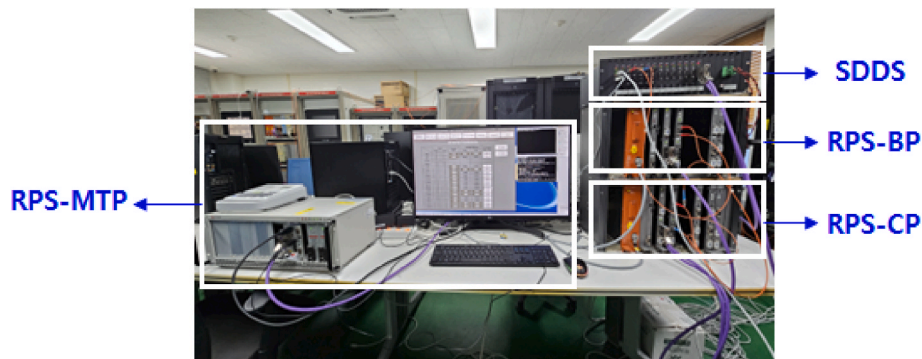


Fig. 12. Configuration of the cyber threat detection system test device linked with the RPS testbed.

tailored to the practical operating environment of NPPs. Notably, the proposed non-invasive detection technique operates independently without altering or impacting the active control system, demonstrating higher practical applicability than Host-based Intrusion Detection Systems (HIDSs).

6. Conclusion

This study proposed a process information-driven cyber threat detection methodology tailored for I&C systems in NPPs. Unlike conventional network traffic-based intrusion detection systems, the proposed approach employs real-time process data analysis to detect cyber threats in safety-critical digital control environments. By analyzing process variables and verifying them against expected operational logic, this method enhances detection accuracy while reducing false positives.

To validate the proposed methodology, a cyber threat detection algorithm was developed and tested on the HPP Trip function of the RPS. Simulation-based evaluations confirmed the effectiveness of detecting unauthorized modifications to the trip setpoint, which could otherwise result in reactor trip failure. Furthermore, experimental validation using the SDDS linked to an APR-1400 RPS testbed demonstrated the system's capability to reliably detect anomalies due to setpoint tampering and logic alterations in safety-grade PLC controllers.

Although the proposed methodology advances NPP cybersecurity, it has certain limitations. This study primarily focused on the HPP Trip function within the RPS, and further research is required to assess its applicability to other control functions such as the NSSS control systems. Additionally, the effectiveness of the detection algorithm should be quantitatively evaluated in terms of false positive rates, false negative rates, detection latency, and computational efficiency. Another critical challenge is the regulatory approval required for integrating such

detection mechanisms into operational nuclear facilities.

Future research will aim to extend the proposed detection framework to various control subsystems and optimizing the algorithm. Moreover, efforts will focus on ensuring compliance with nuclear safety regulations and industrial cybersecurity standards, thereby facilitating the practical implementation of the proposed approach in actual NPP environments.

A key contribution of this study is the development of a non-intrusive and process-based cyber threat detection system deployable in a safety-critical nuclear control environment without interfering with system operations. Unlike host-based detection techniques, which require direct access to control system hardware and software, the proposed method enables real-time monitoring while preserving system integrity and operational stability. These findings underscore the potential to enhance the cybersecurity resilience of NPPs against sophisticated cyber threats, ultimately contributing to the safe and reliable operation of digital control systems in NPPs.

CRediT authorship contribution statement

Jae Hwan Kim: Writing – original draft. **Kwang Seop Son:** Methodology, Conceptualization. **Jae Gu Song:** Resources, Investigation, Conceptualization. **Yong Gu Lee:** Software, Investigation. **Young Jun Lee:** Validation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

2024-08-19 11:15:25.210	CTDA	Threat	1BP_CHA_R1_HPP_threat_detect
2024-08-19 11:15:28.210	CTDA	Threat	1BP_CHA_R1_HPP_threat_detect
2024-08-19 11:25:25.211	CTDA	Threat	1BP_CHA_R1_LPT_PRETRIP_comp
2024-08-19 11:25:25.211	CTDA	Threat	1BP_CHA_R1_LPT_TRIP_comp
2024-08-19 11:25:25.212	CTDA	Threat	1BP_CHA_R1_LPT_threat_detect
2024-08-19 11:35:25.212	CTDA	Threat	1BP_CHA_R1_LPT_threat_detect
2024-08-19 11:40:25.213	CTDA	Threat	1BP_CHA_R1_LSG[11]_PRETRIP_comp
2024-08-19 11:40:25.213	CTDA	Threat	1BP_CHA_R1_LSG[11]_TRIP_comp
2024-08-19 11:40:25.214	CTDA	Threat	1BP_CHA_R1_LSG[11]_threat_detect
2024-08-19 11:42:25.214	CTDA	Threat	1BP_CHA_R1_LSG[11]_threat_detect
2024-08-19 11:50:25.217	CTDA	Threat	1BP_CHA_R1_LSG[11]_PRETRIP_comp
2024-08-19 11:50:25.217	CTDA	Threat	1BP_CHA_R1_LSG[11]_TRIP_comp
2024-08-19 11:53:11.654	CTDA	Threat	1BP_CHA_R1_HSG[11]_threat_detect
2024-08-19 11:57:11.654	CTDA	Threat	1BP_CHA_R1_HSG[11]_threat_detect
2024-08-19 13:15:25.220	CTDA	Threat	1BP_CHA_R1_LSG[11]_TRIP_comp
2024-08-19 13:15:25.221	CTDA	Threat	1BP_CHA_R1_LSG[11]_TRIP_comp
2024-08-19 13:15:25.221	CTDA	Threat	1BP_CHA_R1_LSG[11]_threat_detect
2024-08-19 13:15:25.221	CTDA	Threat	1BP_CHA_R1_LSG[11]_threat_detect
2024-08-19 13:18:51.573	CTDA	Threat	1BP_CHA_R1_HCP_threat_detect
2024-08-19 13:18:51.573	CTDA	Threat	1BP_CHA_R1_HCP_threat_detect
2024-08-19 13:21:19.607	CTDA	Threat	1BP_CHA_R1_HCP_threat_detect
2024-08-19 13:21:19.608	CTDA	Threat	1BP_CHA_R1_HHCP_threat_detect
2024-08-19 13:23:19.608	CTDA	Threat	1BP_CHA_R1_HHCP_threat_detect
2024-08-19 13:25:29.609	CTDA	Threat	1BP_CHA_R1_LACTIF_threat_detect
2024-08-19 13:25:49.609	CTDA	Threat	1BP_CHA_R1_LACTIF_threat_detect
2024-08-19 13:35:25.215	CTDA	Threat	1CP_CHA_R1_VOP_threat_detect
2024-08-19 13:35:35.215	CTDA	Threat	1CP_CHA_R1_VOP_threat_detect
2024-08-19 13:35:38.415	CTDA	Threat	1CP_CHA_R1_HLP_threat_detect
2024-08-19 13:35:45.221	CTDA	Threat	1CP_CHA_R1_HLP_threat_detect
2024-08-19 14:03:16.567	CTDA	Threat	1CP_CHA_R1_HLPD_TRIP_comp
2024-08-19 14:03:16.613	CTDA	Threat	1CP_CHA_R1_HLPD_threat_detect
2024-08-19 14:05:26.413	CTDA	Threat	1CP_CHA_R1_HLPD_threat_detect
2024-08-19 14:25:24.413	CTDA	Threat	1CP_CHA_R1_LDNDR_threat_detect
2024-08-19 14:35:52.113	CTDA	Threat	1CP_CHA_R1_LDNDR_threat_detect
2024-08-19 14:48:34.213	CTDA	Threat	1CP_CHA_R1_HPP_threat_detect
2024-08-19 14:55:26.211	CTDA	Threat	1CP_CHA_R1_HPP_threat_detect
2024-08-19 14:56:13.313	CTDA	Threat	1CP_CHA_R1_LPP_threat_detect
2024-08-19 15:05:11.335	CTDA	Threat	1CP_CHA_R1_LPP_threat_detect
2024-08-19 15:08:25.113	CTDA	Threat	1CP_CHA_R1_LSG[11]_threat_detect
2024-08-19 15:15:22.111	CTDA	Threat	1CP_CHA_R1_LSG[11]_threat_detect

Fig. 13. Detection results log data from the actual cyber threat detection system.

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